QMOF-Rec: Revised Supplementary Material

Authoritative final protocol and reviewer experiments, 19 September 2026

Provenance and interpretation

The executable repository is `qmof-rec`, commit `10082124cf81a90ce40b4bd6306862416a46d2e6`. Source hashes, raw candidate pools, rankings, and reproduction comparisons accompany the artifacts. Twenty-one mask-aware regression tests pass. All seven primary methods reproduce Rel@5, NDCG@5, and diversity at the original four-decimal precision. There are five scenarios and ten seeds (42–51), not 50 independent scientific queries. Density is observed for 20,372 records, band gap for 10,810, and void fraction for none. Historical artifacts absent from this checkout are not asserted to have been reproduced.

Complete scenario-level ranking results

Metrics are computed on the same pool of 100 for each query/seed. Values below are means over ten seeds. MMR (top-5) exactly matches the original MMR top five in all 50 runs. DE uses the same representation and evaluation but a fixed list objective and 1,800 portfolio evaluations. Relevance/NDCG are aligned utility diagnostics, not independent labels.

Table S1: Complete primary-method results by scenario. SD describes diversity across seeds.

Scenario	Method	Rel@5	NDCG	Diversity	SD	ms
CO_2	WeightedSum	0.7599	1.0000	0.0163	0.0000	0.05
CO_2	LEA	0.7599	1.0000	0.0288	0.0026	85.66
CO_2	TOPSIS	0.7599	1.0000	0.0163	0.0000	1.18
CO_2	ParetoCrowding	0.7599	1.0000	0.0297	0.0008	224.13
CO_2	MMR (top-5)	0.7599	1.0000	0.0333	0.0011	12.65
CO_2	Random	0.7377	0.9644	0.0189	0.0043	0.05
CO_2	SemanticOnly	0.7265	0.9395	0.0228	0.0035	0.04
CO_2	DE portfolio	0.7599	1.0000	0.0349	0.0014	70.24
Photocatalysis	WeightedSum	0.8752	1.0000	0.0150	0.0000	0.05
Photocatalysis	LEA	0.8752	1.0000	0.0150	0.0000	85.15
Photocatalysis	TOPSIS	0.8752	1.0000	0.0150	0.0000	1.13
Photocatalysis	ParetoCrowding	0.8752	1.0000	0.0150	0.0000	225.57
Photocatalysis	MMR (top-5)	0.8752	1.0000	0.0150	0.0000	12.71
Photocatalysis	Random	0.7624	0.8329	0.0232	0.0061	0.05
Photocatalysis	SemanticOnly	0.7552	0.8245	0.0213	0.0049	0.04
Photocatalysis	DE portfolio	0.8752	1.0000	0.0150	0.0000	70.36
Lightweight	WeightedSum	0.9174	1.0000	0.0165	0.0000	0.05
Lightweight	LEA	0.9174	1.0000	0.0193	0.0022	87.84
Lightweight	TOPSIS	0.9174	1.0000	0.0165	0.0000	1.11
Lightweight	ParetoCrowding	0.9013	0.9761	0.0378	0.0055	226.89
Lightweight	MMR (top-5)	0.9174	1.0000	0.0204	0.0006	12.67
Lightweight	Random	0.8967	0.9692	0.0266	0.0054	0.05
Lightweight	SemanticOnly	0.8944	0.9665	0.0260	0.0032	0.04
Lightweight	DE portfolio	0.9174	1.0000	0.0194	0.0011	68.89
Balanced	WeightedSum	0.8979	1.0000	0.0165	0.0000	0.05
Balanced	LEA	0.8979	1.0000	0.0194	0.0010	92.43
Balanced	TOPSIS	0.8979	1.0000	0.0165	0.0000	1.13
Balanced	ParetoCrowding	0.8979	1.0000	0.0185	0.0000	220.62
Balanced	MMR (top-5)	0.8979	1.0000	0.0204	0.0006	12.73
Balanced	Random	0.7647	0.8062	0.0256	0.0044	0.05
Balanced	SemanticOnly	0.7548	0.7980	0.0250	0.0051	0.04
Balanced	DE portfolio	0.8979	1.0000	0.0199	0.0012	68.92
Insulating	WeightedSum	0.7893	1.0000	0.0293	0.0000	0.05
Insulating	LEA	0.7893	1.0000	0.0294	0.0014	85.25
Insulating	TOPSIS	0.7893	1.0000	0.0293	0.0000	1.08
Insulating	ParetoCrowding	0.7893	1.0000	0.0303	0.0006	226.21

Scenario	Method	Rel@5	NDCG	Diversity	SD	ms
Insulating	MMR (top-5)	0.7893	1.0000	0.0316	0.0018	12.77
Insulating	Random	0.6860	0.8320	0.0241	0.0061	0.05
Insulating	SemanticOnly	0.6693	0.8106	0.0249	0.0054	0.04
Insulating	DE portfolio	0.7893	1.0000	0.0317	0.0011	68.51

Joint gamma and redundancy sensitivity

All 35 settings are rerun on all 50 pools. Changed sets and orders are counted relative to $(\gamma, \lambda) = (0.10, 0.05)$. The true no-diversity ablation is $(0, 0)$; setting only gamma to zero retains a diversity-dependent penalty. Full per-scenario, per-seed metrics and selected IDs are saved in `artifacts/reviewer_revision/sensitivity.csv`.

Table S2: Joint sensitivity; identical rounded metrics can conceal different selected sets.

γ	λ	Rel@5	NDCG	Diversity	Sets /50	Orders /50
0	0	0.8479	1.0000	0.0187	40	50
0	0.025	0.8479	1.0000	0.0224	0	0
0	0.05	0.8479	1.0000	0.0224	0	0
0	0.1	0.8479	1.0000	0.0224	0	0
0	0.2	0.8282	0.9707	0.0271	30	30
0.025	0	0.8479	1.0000	0.0215	39	50
0.025	0.025	0.8479	1.0000	0.0224	0	0
0.025	0.05	0.8479	1.0000	0.0224	0	0
0.025	0.1	0.8479	1.0000	0.0224	0	0
0.025	0.2	0.8282	0.9707	0.0271	30	30
0.05	0	0.8479	1.0000	0.0215	39	50
0.05	0.025	0.8479	1.0000	0.0224	0	0
0.05	0.05	0.8479	1.0000	0.0224	0	0
0.05	0.1	0.8479	1.0000	0.0224	0	0
0.05	0.2	0.8282	0.9707	0.0271	30	30
0.1	0	0.8479	1.0000	0.0215	39	50
0.1	0.025	0.8479	1.0000	0.0224	0	0
0.1	0.05	0.8479	1.0000	0.0224	0	0
0.1	0.1	0.8479	1.0000	0.0224	0	0
0.1	0.2	0.8282	0.9707	0.0271	30	30
0.2	0	0.8479	1.0000	0.0215	39	50
0.2	0.025	0.8479	1.0000	0.0224	0	0
0.2	0.05	0.8479	1.0000	0.0224	0	0
0.2	0.1	0.8479	1.0000	0.0224	0	0
0.2	0.2	0.8282	0.9707	0.0271	30	30
0.4	0	0.8479	1.0000	0.0215	39	50
0.4	0.025	0.8479	1.0000	0.0224	0	0
0.4	0.05	0.8479	1.0000	0.0224	0	0
0.4	0.1	0.8479	1.0000	0.0224	0	0
0.4	0.2	0.8282	0.9707	0.0271	30	30
0.8	0	0.8478	0.9999	0.0218	39	50
0.8	0.025	0.8478	0.9999	0.0219	13	35
0.8	0.05	0.8478	0.9999	0.0225	1	1
0.8	0.1	0.8477	0.9998	0.0226	1	1
0.8	0.2	0.8278	0.9703	0.0274	32	32

Runtime dispersion and scaling

Reranking times are serial wall-clock measurements; they exclude shared catalog scoring, retrieval, networking, and explanations. Table S3 summarizes 50 query/seed runs. The original MMR full-list row is an audit comparator, not the matched top-five implementation. Scaling uses 45 measurements per method/size (five queries, three seeds, three repeats), with population 30 and 60 LEA iterations fixed.

Table S3: Primary-method runtime dispersion; MMR without the top-5 suffix computes the full 100 ranks.

Method	Mean ms	SD ms	Minimum	Maximum
WeightedSum	0.051	0.005	0.044	0.068
LEA	87.264	10.326	81.479	153.210
TOPSIS	1.125	0.111	0.993	1.567
ParetoCrowding	224.682	11.216	207.015	291.763
MMR (top-5)	12.704	0.275	12.109	13.462
Random	0.051	0.004	0.045	0.065
SemanticOnly	0.039	0.009	0.030	0.071
DE portfolio	69.386	2.253	67.609	79.996
MMR	2209.671	108.106	2068.668	2597.575

Table S4: Post-retrieval scalability at fixed top-five output.

Pool size	Method	Measurements	Mean ms	SD ms
50	WeightedSum	45	0.029	0.015
50	TOPSIS	45	0.501	0.055
50	LEA	45	76.514	2.868
50	MMR (top-5)	45	6.056	0.137
100	WeightedSum	45	0.057	0.025
100	TOPSIS	45	1.000	0.103
100	LEA	45	82.996	0.604
100	MMR (top-5)	45	12.611	0.298
200	WeightedSum	45	0.120	0.047
200	TOPSIS	45	2.010	0.167
200	LEA	45	97.948	1.403
200	MMR (top-5)	45	25.611	0.402
500	WeightedSum	45	0.315	0.102
500	TOPSIS	45	4.990	0.318
500	LEA	45	140.972	7.039
500	MMR (top-5)	45	65.155	0.847
1000	WeightedSum	45	0.679	0.184
1000	TOPSIS	45	10.189	0.542
1000	LEA	45	229.754	76.462
1000	MMR (top-5)	45	137.550	30.063

Descriptor-level screening sanity check

Density is in g/cm^3 and band gap in eV. Means average the selected observed descriptors within each seed, then average across seeds with defined means; all-missing seed means are excluded, not zero-filled. A dash denotes an undefined summary. Catalog and pool are reference populations, not experimental control groups. Values are not adsorption capacity or evidence of synthesis feasibility. Missingness is reported separately; thresholds are descriptive. Formula and text heuristics can dominate the retained scenario definitions.

Table S5: Observed descriptor means and band-gap completeness; the threshold percentage uses observed gaps as denominator.

Scenario	Group	Density	Gap	Gap missing %	Gap $\geq$3.5 %
CO_2	catalog	1.696	3.858	46.9	64.5
CO_2	candidate pool	1.619	3.325	0.0	33.0
CO_2	Random	1.639	3.213	0.0	24.0
CO_2	WeightedSum	1.537	3.070	0.0	0.0
CO_2	LEA	1.640	2.116	0.0	0.0
Photocatalysis	catalog	1.696	3.858	46.9	64.5

Scenario	Group	Density	Gap	Gap missing %	Gap ≥3.5 %
Photocatalysis	candidate pool	1.647	2.922	0.0	0.0
Photocatalysis	Random	1.662	2.912	0.0	0.0
Photocatalysis	WeightedSum	0.870	2.804	0.0	0.0
Photocatalysis	LEA	0.870	2.804	0.0	0.0
Lightweight	catalog	1.696	3.858	46.9	64.5
Lightweight	candidate pool	0.688	2.873	93.0	0.0
Lightweight	Random	0.683	2.758	90.0	0.0
Lightweight	WeightedSum	0.859	2.693	0.0	0.0
Lightweight	LEA	0.890	2.911	0.0	0.0
Balanced	catalog	1.696	3.858	46.9	64.5
Balanced	candidate pool	1.518	3.021	0.0	12.0
Balanced	Random	1.609	2.947	0.0	4.0
Balanced	WeightedSum	0.859	2.693	0.0	0.0
Balanced	LEA	0.871	2.801	0.0	0.0
Insulating	catalog	1.696	3.858	46.9	64.5
Insulating	candidate pool	1.594	4.307	0.0	100.0
Insulating	Random	1.537	4.071	0.0	100.0
Insulating	WeightedSum	0.771	4.211	0.0	100.0
Insulating	LEA	0.810	4.311	0.0	100.0

Ablation scope and advanced optimizer

Zero-imputed WeightedSum changes the missing-score convention on the same pools, with evaluation using the original masked relevance. If its top five are unchanged, that is a null effect in this pool, not a demonstration that masking is unnecessary. Uniform-pool LEA replaces candidate generation with random sampling from the catalog and therefore changes the feasible set and NDCG ideal. Its scores cannot be read as a same-pool optimizer comparison. Proxy S/A variants use 85% relevance and 15% engineered proxy, not trained GNN output.

The DE portfolio implementation uses SciPy 1.13.1, `best1bin`, differential weight dithering 0.5–1, crossover 0.7, immediate updating, no polishing, 30 members, and 59 generations after initialization. Random keys select exactly five unique IDs. One population member encodes WeightedSum, with 29 random members. Its fixed list objective is $\overline{R} + 0.1\overline{B} + 0.1D - 0.05e^{-D}$. Cached pairwise distance construction is included in timing; outputs are ordered by relevance. Tests verify scalar/vectorized masked-distance agreement, unique outputs, 1,800 evaluations, and utility no worse than the initial WeightedSum portfolio. Equal nominal search budgets do not equate candidate-fitness evaluations with whole-portfolio evaluations, nor establish identical optimization problems.

Implementation consistency and missingness

The final protocol and configuration metadata specify eight formula coordinates, including heteroatom fraction. The executed hybrid vector contains two physical, eight formula, and two proxy dimensions, with block scaling $\sqrt{0.55/2}$, $\sqrt{0.30/8}$, and $\sqrt{0.15/2}$ before masked RMS distance. The missing physical gap coordinate is excluded, but proxy A has an internal zero fallback. This retained heuristic is disclosed rather than silently changed during reproduction. Objective remapping and hybrid diversity remain distinct. LEA retains best-ever candidate fitness across changing selected contexts; its trace is not convergence of a single fixed portfolio objective.

Explanation audit

The runner emits 30 templates from saved seed-42 LEA selections. Groundedness uses identifier presence and limitation awareness uses phrases. Metadata consistency, nominal explanation-quality scores, and violation flags are assigned, rather than assessed by an independent evaluator. The former

5/5 and zero-hallucination claims are therefore excluded as evidence of explanation quality. No live LLM or expert-human benchmark has been run.

Reproduction commands and files

Run from the authoritative repository:

```
PYTHONPATH=backend python3 -m pytest -q backend/tests
python3 scripts/run_final_diversity_aware_masked_protocol.py
python3 scripts/run_reviewer_revision.py
python3 scripts/build_reviewer_revision.py
```

The baseline configuration is `configs/final_diversity_aware_masked_protocol.json`. New raw results, manifests, figures, and source snapshots are in `artifacts/reviewer_revision/`. The manuscript builder writes only the designated revision workspace and retains input sources under the executable repository's source snapshots. The versioned reproducibility archive `qmof-rec-v1.1-COMMT-revision.tar.gz` contains the frozen evidence, executable ranking code, regression tests, and a SHA-256 manifest.

Historical interface illustrations

The following screenshots are retained as interface illustrations. They are not new experiments, physical validation, or an independently assessed LLM benchmark.

CO_2 Adsorption Scenario

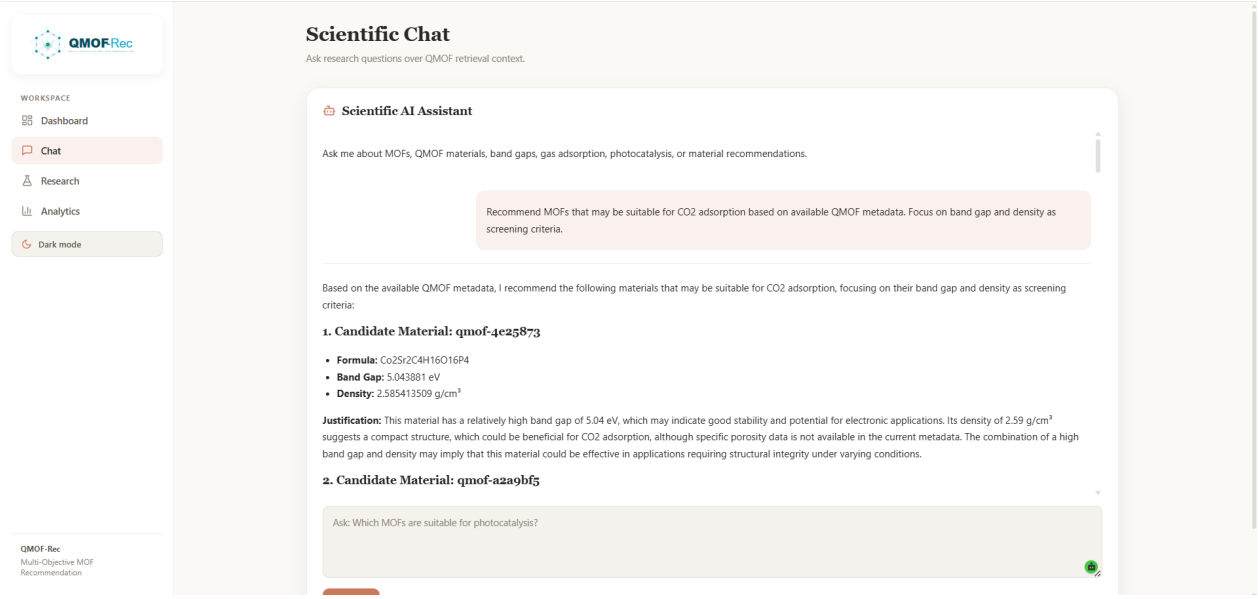


Figure S1: CO_2 adsorption scenario. The assistant recommends candidates based on available metadata and states that porosity and CO_2 uptake data are unavailable. Void fraction and adsorption uptake are unavailable in the current metadata.

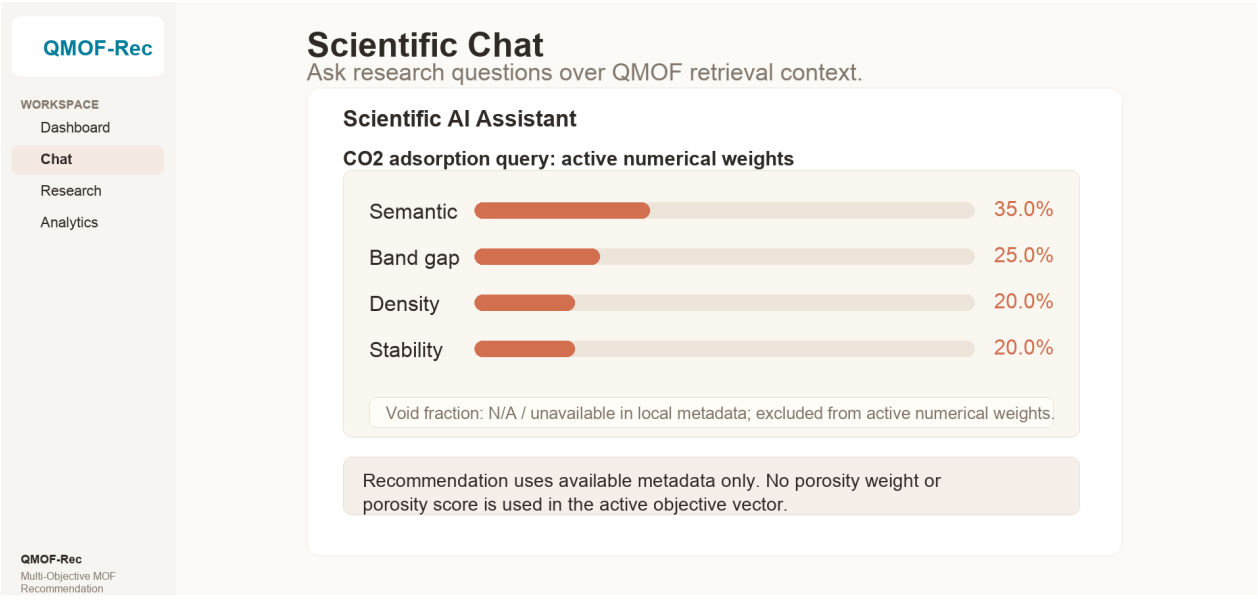


Figure S2: CO_2 adsorption scenario showing active numerical weights after excluding unavailable void-fraction information. Void fraction is displayed only as unavailable metadata and is not used as an active numerical objective, weight, or score.

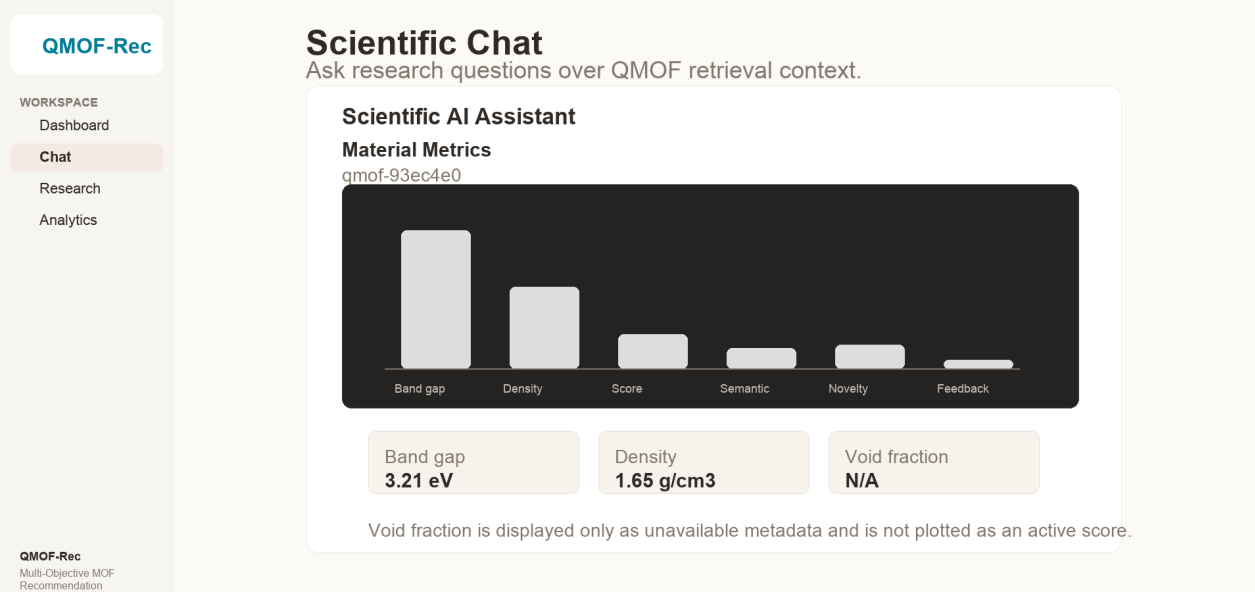


Figure S3: CO_2 adsorption scenario showing material metrics for a candidate record. In the revised implementation, void fraction is displayed only as unavailable metadata and is excluded from active numerical ranking.

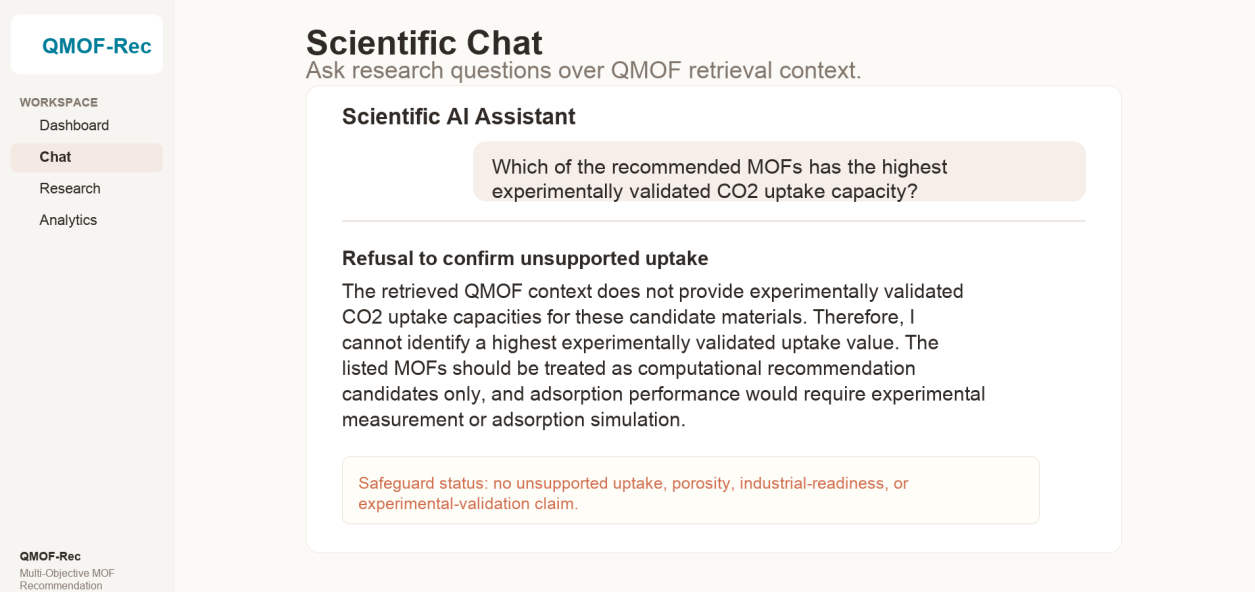


Figure S4: CO_2 adsorption scenario testing unsupported uptake claims. The assistant declines to identify experimentally validated CO_2 uptake because the retrieved context contains no experimentally validated uptake data.

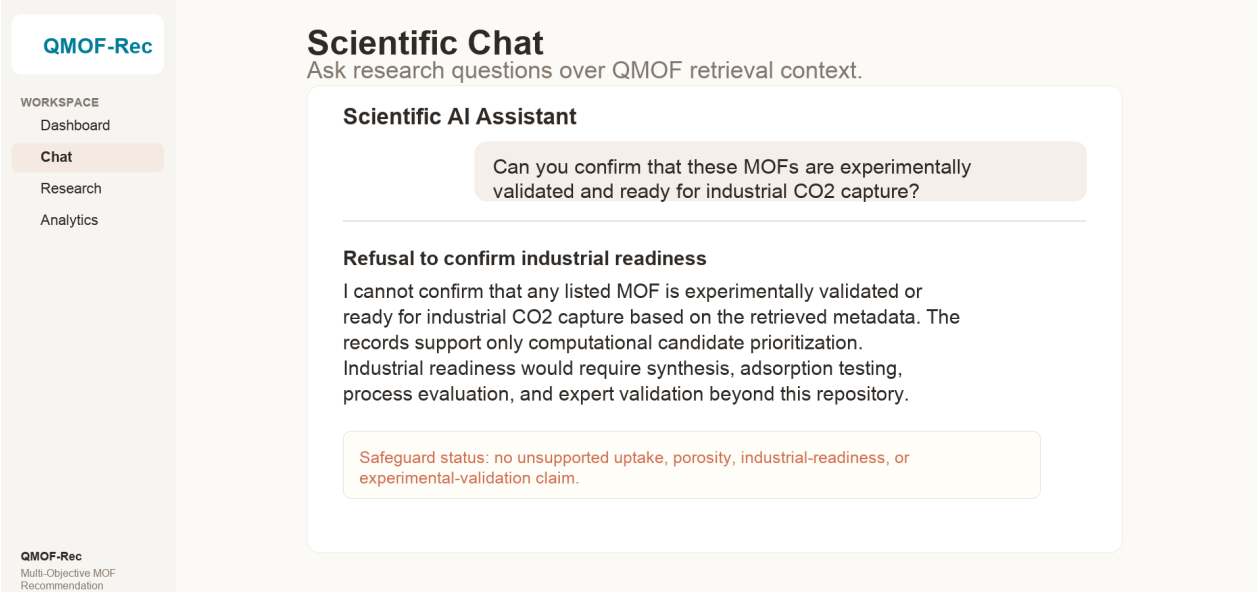
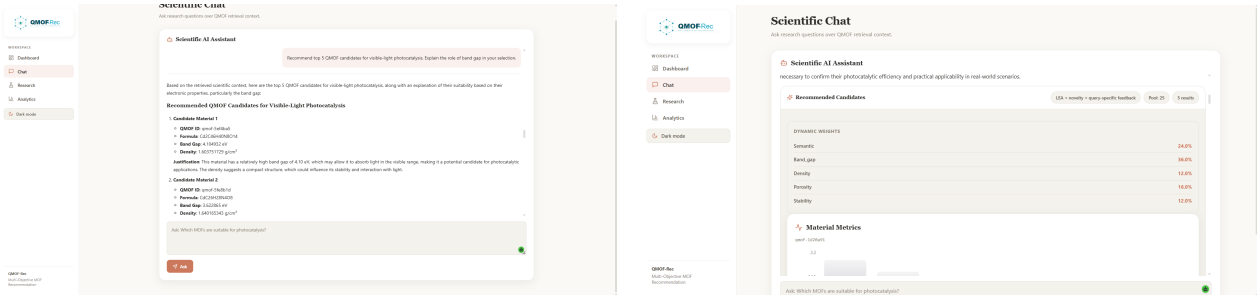


Figure S5: CO_2 adsorption scenario testing industrial-readiness claims. The assistant treats the listed MOFs as recommendation candidates requiring expert follow-up, not as experimentally validated capture materials.

Photocatalysis Scenario

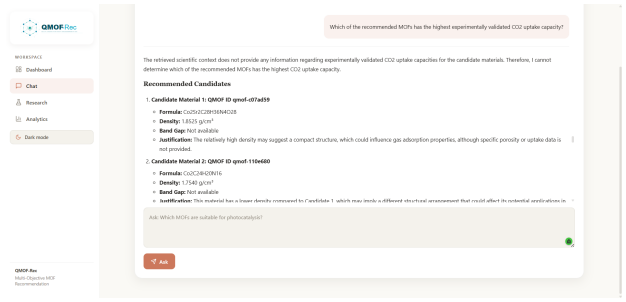


(a) Top-5 photocatalysis candidates with band-gap-oriented justifications.

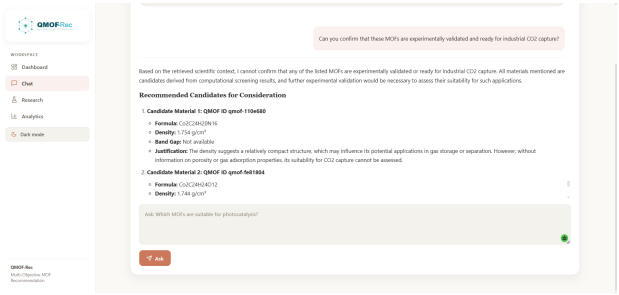
(b) Dynamic objective weights for a photocatalysis-oriented query.

Figure S6: Photocatalysis scenario screenshots. These illustrate answer structure and interface behavior rather than physical validation.

Missing-Information Trap



(a) Refusal to confirm experimentally validated CO_2 uptake.



(b) Refusal to confirm industrial readiness.

Figure S7: Missing-information trap screenshots. The assistant declines to fabricate experimental validation claims.

Lightweight Gas Storage

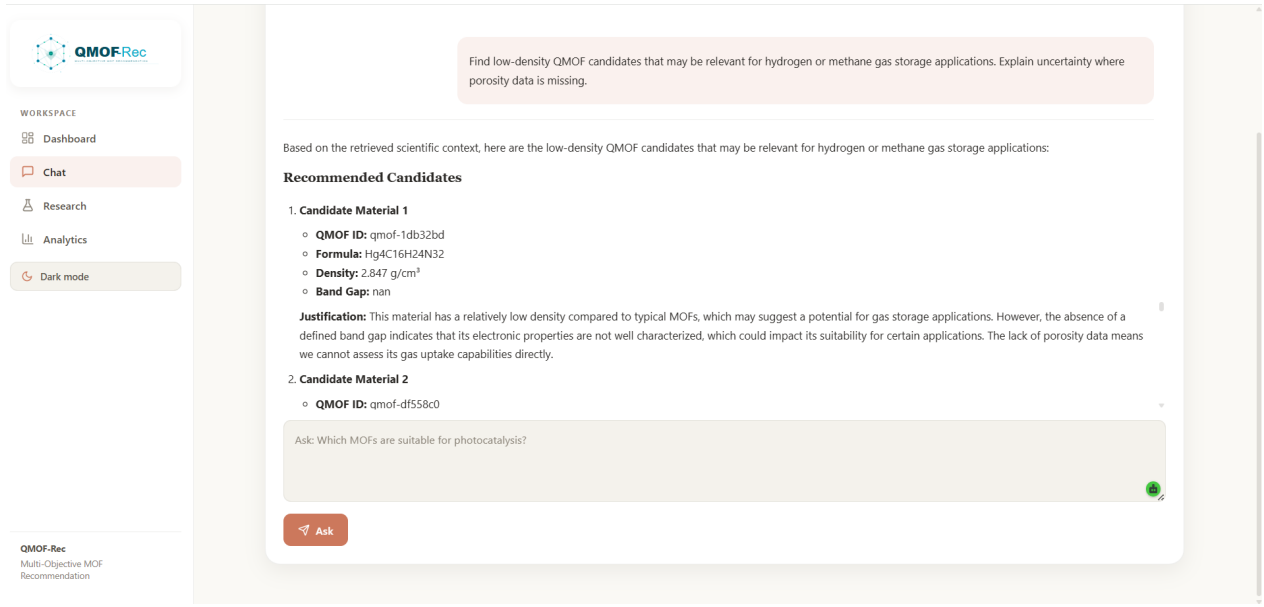


Figure S8: Lightweight gas-storage scenario. Candidate recommendations are based on available metadata such as density; void fraction, porosity, and gas uptake data are unavailable, so conclusions are preliminary.

Ranking Logic

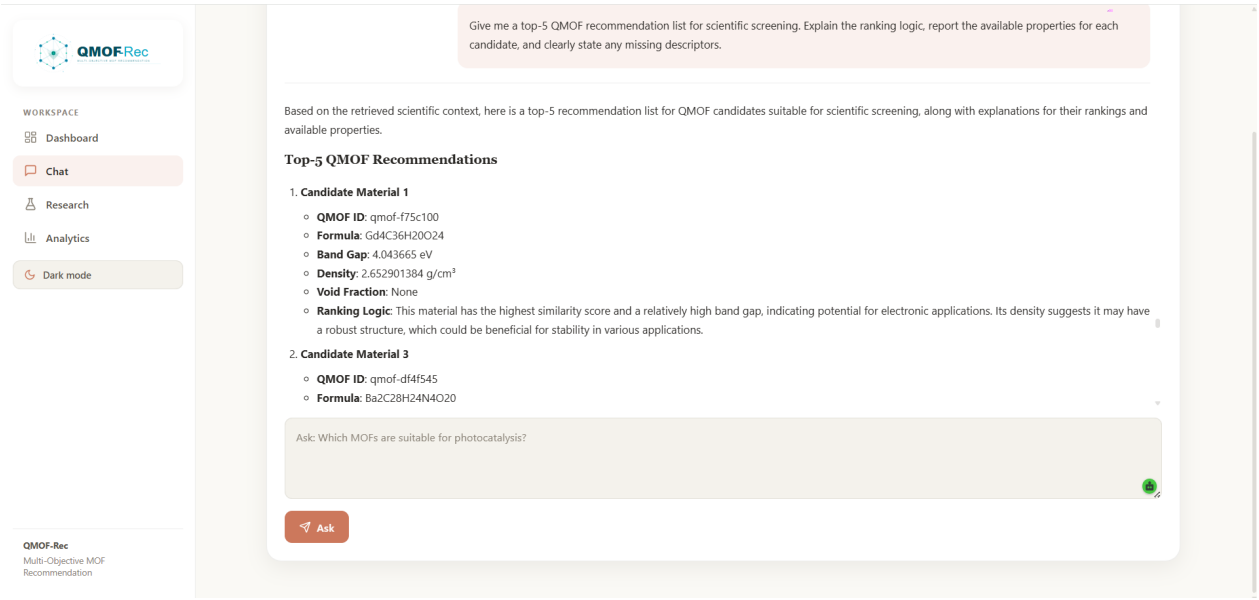


Figure S9: Ranking-logic scenario. The assistant provides a structured recommendation list with explicit ranking rationale and missing-descriptor reporting.

Interface Example: Candidate Diversity

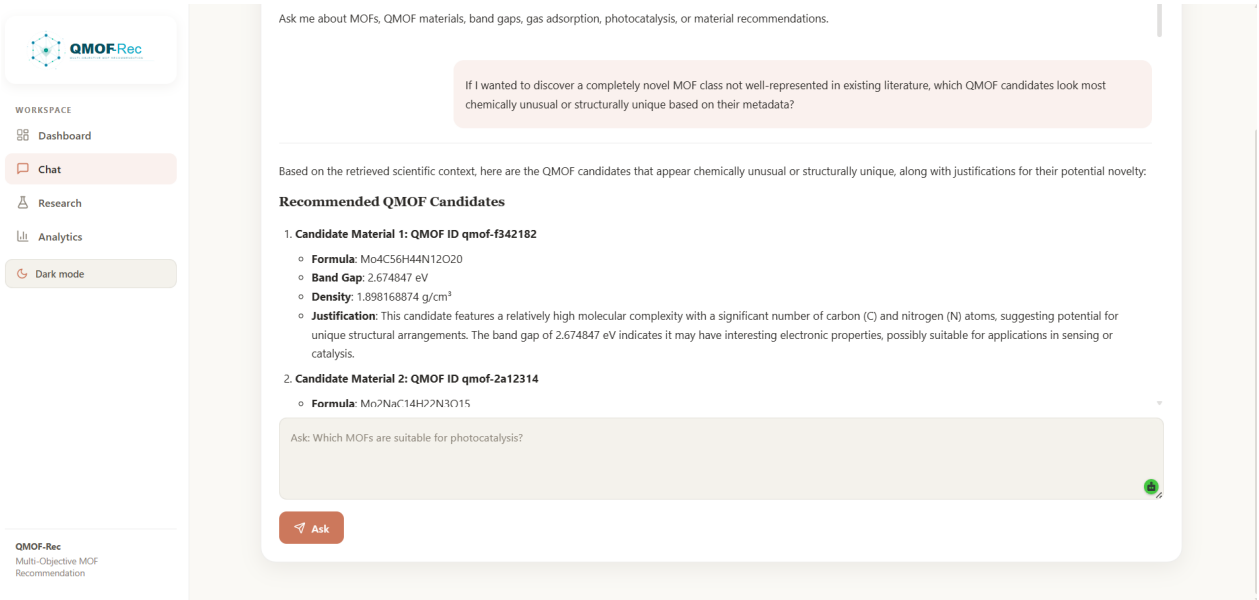


Figure S10: Candidate-diversity interface scenario. The response is grounded in available metadata and should not be interpreted as experimental, adsorption, porosity, or structure-level validation.